

Project Title: Digital media consumption and market trends

Author: Maverick Luke

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Imports and Constants

Introduction

The streaming media landscape has evolved dramatically in recent years, with platforms like Netflix and Disney+ competing intensely for subscribers. Original content has become a primary differentiator in this competition, with platforms investing billions in developing exclusive shows and movies. Meanwhile, social media platforms like Reddit serve as immediate indicators of audience reception, potentially signaling a release's success or failure. This intersection of streaming content strategies and social media engagement raises an important question for investors: How do major content releases on these platforms affect their stock market performance, and can social media metrics serve as leading indicators for these market movements?

Research Question

This project addresses the question: **How do major streaming content releases on Netflix and Disney+ affect their respective stock prices in the short term, and does social media engagement correlate with these market movements?**

Specifically, we analyze:

1. Whether stock prices show abnormal returns in the 3-day window following content releases
2. Whether Reddit engagement metrics correlate with stock price movements
3. Is the sentiment of Reddit comments about a movie (positive/negative) associated with upward/downward movement in stock price?
4. Is there a lagged relationship between Reddit engagement and provider stock returns (e.g., 1-day, 3-day, 7-day lag)?

Methods

The study combines data from three distinct sources to examine relationships between streaming content releases, stock market performance, and social media engagement:

- Streaming Content Data: **The Movie Database (TMDB) API** was used to collect information about Netflix and Disney+ releases, including release dates, popularity scores, and genres. Both TV shows and movies were included to ensure a sufficient sample size.
- Stock Market Data: Financial data for Netflix (NFLX), Disney (DIS), and the S&P 500 index was collected using the Yahoo Finance API via the Python **yfinance** package. Daily closing prices, trading volume, and other metrics were gathered for the period from September 2024 to March 2025.
- Social Media Engagement Data: Reddit's API was accessed using the Python **praw** library to collect post and comment data related to each content release. Data was collected from relevant subreddits including r/television, r/netflix, r/DisneyPlus, r/streaming.

API Implementation

(Skip to Read existing CSV files if the dataframes are available in CSVs)

The following code demonstrates how data was collected from each API source: (Note that the API calls take some time and requires API keys (for TMDB) and client id and client secret (for Reddit praw). To improve efficiency, I've saved the datasets as CSV files for quicker retrieval. The retrieval code is located right below the API calls functions)

Content Data Preprocessing

The data structures for TV shows and movies were standardized to ensure consistency. For TV shows, the field `first_air_date` was renamed to `release_date` to align with the naming convention used for movies. Additionally, a `content_type` field was introduced to clearly distinguish between movies and TV shows, making it easier to organize the data. This standardization simplifies comparisons and ensures a unified format across different types of content. Genre information is retrieved and one-hot encoded to enable analysis by genre. Retrieved releases are sorted by popularity and filtered to include only those with at least 1000 votes.

Stock Data Preprocessing

In the stock data preprocessing phase, two key calculations were performed to analyze stock performance: return calculation and market adjustment. First, daily returns were computed as the percentage change in the adjusted closing price from the previous day, providing insight into short-term price movements. To isolate company-specific movements from broader market trends, abnormal returns were calculated. This was achieved by subtracting the market return (represented by the S&P 500 index as the benchmark) from the stock return. The formula used was: **`abnormal_return = stock_return - market_return`**. Finally, event windows were defined to examine the impact of content releases on stock performance. A 3-day post-release window (from the release day to two days after) was specifically analyzed to capture short-term stock reactions to the release of new content.

Social Media Data Processing

Reddit posts were filtered to a 28-day window (14 days before and after each release) to capture audience engagement around content launches. Posts are filtered by relevance and cleaned titles to improve matching. For each post, metadata (e.g., score, upvote ratio, comment count) and top comments are collected. Sentiment analysis is applied to both posts and comments using VADER, and sentiment scores are weighted by engagement metrics. The final output includes summary statistics such as **average sentiment**, **weighted sentiment**, and a custom **engagement score** incorporating upvotes, comments, and controversy.

Data Integration

The processed datasets were merged based on content title and release date. The final integrated dataset included the following key variables:

- Content information: title, release date, platform, popularity, genre, content type
- Stock performance: abnormal returns across different time windows
- Social media engagement: average sentiment, weighted sentiment, engagement score

Tools and Libraries Used

The analysis was conducted using the following Python libraries:

- Data Collection: requests, yfinance, praw
- Data Processing: pandas, numpy
- Sentiment analysis: VADER
- Time Series Handling: datetime

- Visualization: matplotlib, seaborn

We save each dataset into csv files so that anytime we need the data, we can just read from them.

Read existing CSV files

Exploratory Data Analysis

Releases Data:

	id	popularity	vote_average	vote_count	provider_id	Acti Adver
count	4.100000e+02	410.000000	410.000000	410.000000	410.000000	410.00
mean	2.239948e+05	84.410629	7.661332	7576.521951	172.500000	0.12
std	3.243610e+05	57.161458	0.733704	7701.313922	164.700977	0.33
min	1.100000e+01	30.307000	5.359000	1005.000000	8.000000	0.00
25%	1.981000e+03	41.796600	7.184000	2250.000000	8.000000	0.00
50%	4.889100e+04	74.702200	7.779000	4165.000000	172.500000	0.00
75%	3.398460e+05	105.767800	8.239000	10621.000000	337.000000	0.00
max	1.241982e+06	451.814700	8.924000	37003.000000	337.000000	1.00

8 rows × 32 columns



Stock Data:

	Adj Close	Close	High	Low	Open	
count	18521.000000	18521.000000	18521.000000	18521.000000	18521.000000	1.8
mean	816.149174	817.689592	822.894792	811.895670	817.573579	1.7
std	1241.425702	1240.483257	1246.889287	1233.193557	1240.305614	1.8
min	0.372857	0.372857	0.410714	0.346429	0.377857	2.8
25%	27.137035	31.770000	32.238819	31.360001	31.840000	7.5
50%	116.214333	117.849998	119.160004	116.500000	117.940002	1.5
75%	1229.010010	1229.010010	1236.709961	1221.130005	1229.030029	2.4
max	6144.149902	6144.149902	6147.430176	6111.149902	6134.500000	1.7



Abnormal Returns Data:

	id	popularity	vote_average	vote_count	provider_id	Acti Adver
count	3.130000e+02	313.000000	313.000000	313.000000	313.000000	313.00
mean	2.923244e+05	87.163949	7.579236	7025.367412	177.230032	0.14
std	3.436532e+05	56.462701	0.736468	7151.996407	164.695284	0.34
min	9.800000e+01	30.307000	5.897000	1005.000000	8.000000	0.00
25%	3.113200e+04	43.478100	7.084000	2271.000000	8.000000	0.00
50%	1.000880e+05	76.644800	7.622000	3738.000000	337.000000	0.00
75%	4.260630e+05	113.067000	8.213000	10455.000000	337.000000	0.00
max	1.241982e+06	451.814700	8.924000	37003.000000	337.000000	1.00

8 rows × 35 columns



Reddit Engagement Data:

	n_texts	avg_sentiment	std_sentiment	min_sentiment	max_sentiment
count	410.000000	185.000000	185.000000	185.000000	185.000000
mean	22.868293	0.165588	0.527095	-0.815312	0.929378
std	45.291743	0.190671	0.085897	0.247913	0.110835
min	0.000000	-0.563358	0.230730	-0.999900	0.361200
25%	0.000000	0.064827	0.490240	-0.971500	0.897900
50%	0.000000	0.166509	0.531898	-0.889200	0.973500
75%	20.000000	0.285767	0.573069	-0.787700	0.997200
max	281.000000	0.567145	0.715923	0.000000	0.999800

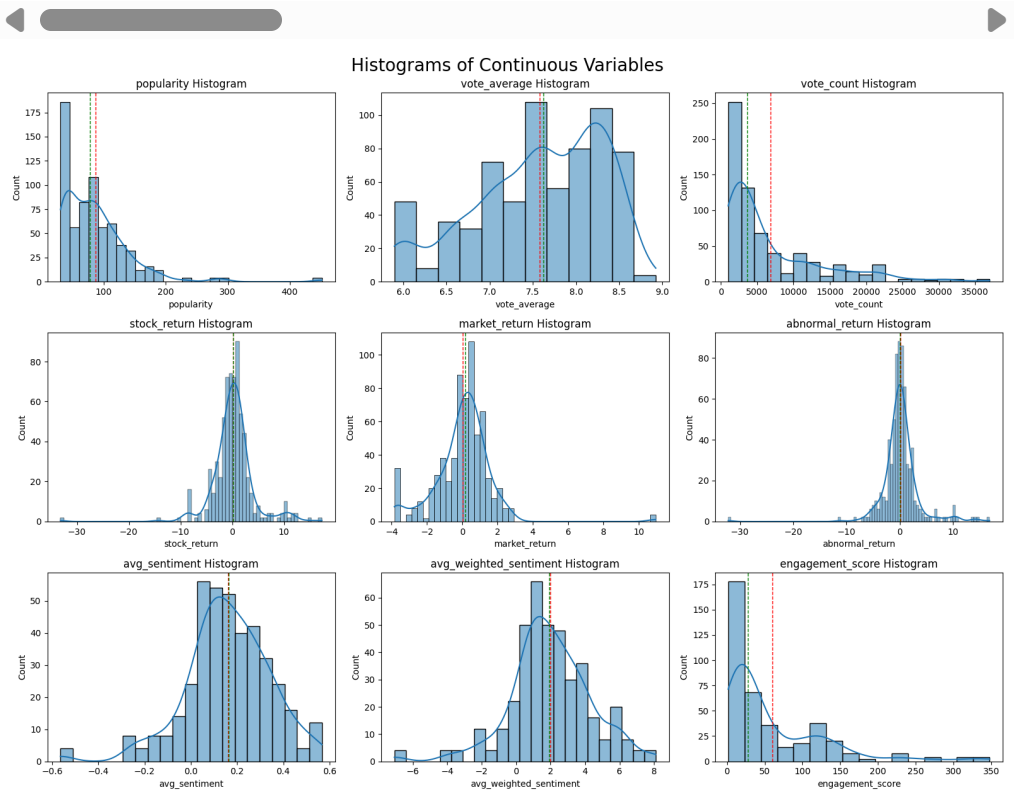


Daily return calculations require both the current and previous day's adjusted close values. Each of our 3 tickers has one row where the previous day's data is unavailable, resulting in a null value for the daily return on that specific row. This is an expected consequence of our finite dataset rather than an error in the data. This is why we remove the rows with nulls in the stock data.

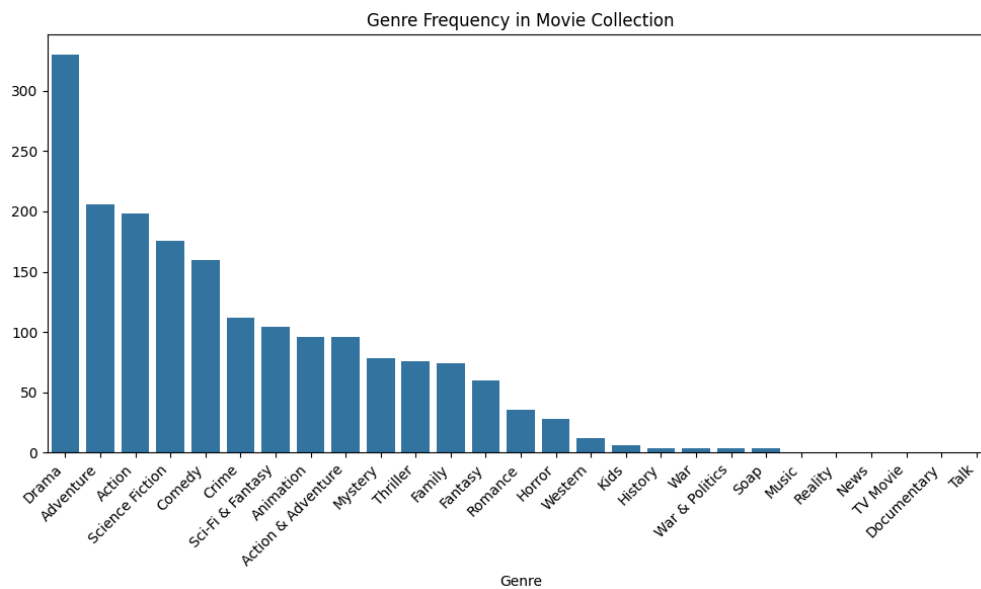
Summary Statistics for the Merged Dataset:

	popularity	vote_average	vote_count	Action & Adventure	Animation	Comedy
count	674.000000	674.000000	674.000000	674.000000	674.000000	674.000000
mean	86.432187	7.576312	6829.899110	0.142433	0.142433	0.237300
std	54.921773	0.738511	6998.357379	0.349753	0.349753	0.425700
min	30.307000	5.897000	1005.000000	0.000000	0.000000	0.000000
25%	43.037400	7.100000	2268.000000	0.000000	0.000000	0.000000
50%	77.385200	7.622000	3593.000000	0.000000	0.000000	0.000000
75%	110.676900	8.210000	10325.000000	0.000000	0.000000	0.000000
max	451.814700	8.924000	37003.000000	1.000000	1.000000	1.000000

8 rows × 42 columns

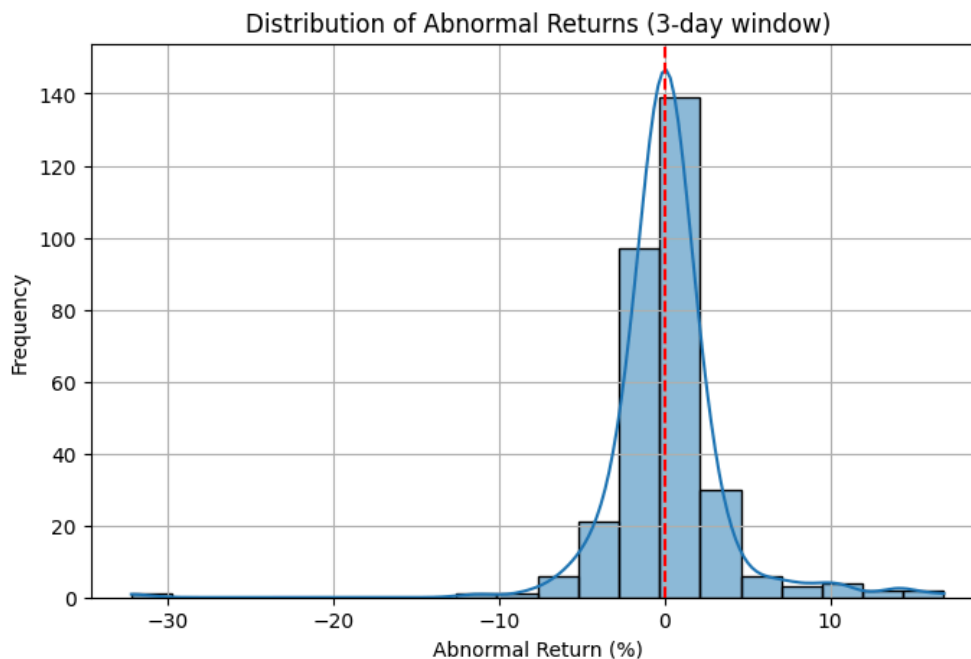


The histograms of the continuous variables reveal several key patterns in their distributions. Most variables—such as popularity, vote_count, engagement_score, and the return-based metrics—exhibit strong right-skewness. In contrast, vote_average, avg_sentiment, and avg_weighted_sentiment appear more symmetrically distributed, with avg_sentiment closely resembling a normal distribution centered around a slightly positive value. Return variables (stock_return, market_return, and abnormal_return) cluster tightly around zero. Overall, the distributions highlight a landscape dominated by a few highly impactful data points against a backdrop of more modest, frequent observations.



The bar chart above displays the frequency distribution of genres across a movie collection, with Drama being the most prevalent genre, followed by Adventure and Action.

Subquestion 1



One-sample t-test for abnormal returns:

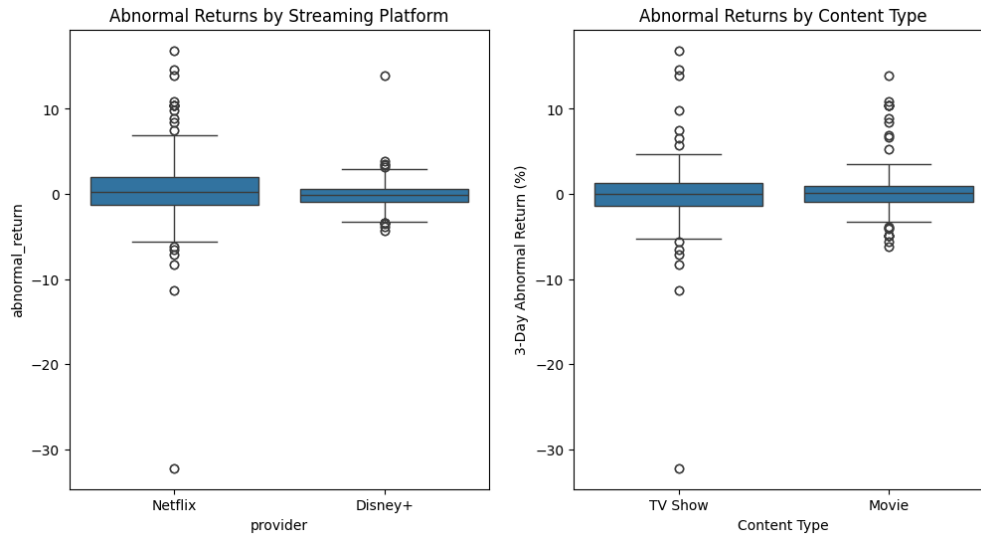
T-statistic: 0.8550

P-value: 0.3932

Based on the figure and the t-test results (t-statistic = 0.855, p-value = 0.393), we fail to reject the null hypothesis. This indicates that the mean abnormal return does not differ significantly from zero. This addresses our first subquestion: there is no statistically significant stock movement, on average, in the 3-day window following a major streaming release.

Subquestion 2

Text(0, 0.5, '3-Day Abnormal Return (%)')



The boxplots compare 3-day abnormal stock returns after content releases. Boxplots showing the distribution of 3-day abnormal returns (%) segmented by streaming platform (Netflix vs. Disney+) and content type (TV Show vs. Movie). Netflix exhibits greater variability and more extreme outliers compared to Disney+. Similarly, TV Shows show slightly wider spread in abnormal returns than Movies, though both content types display a generally normal distribution centered around zero.

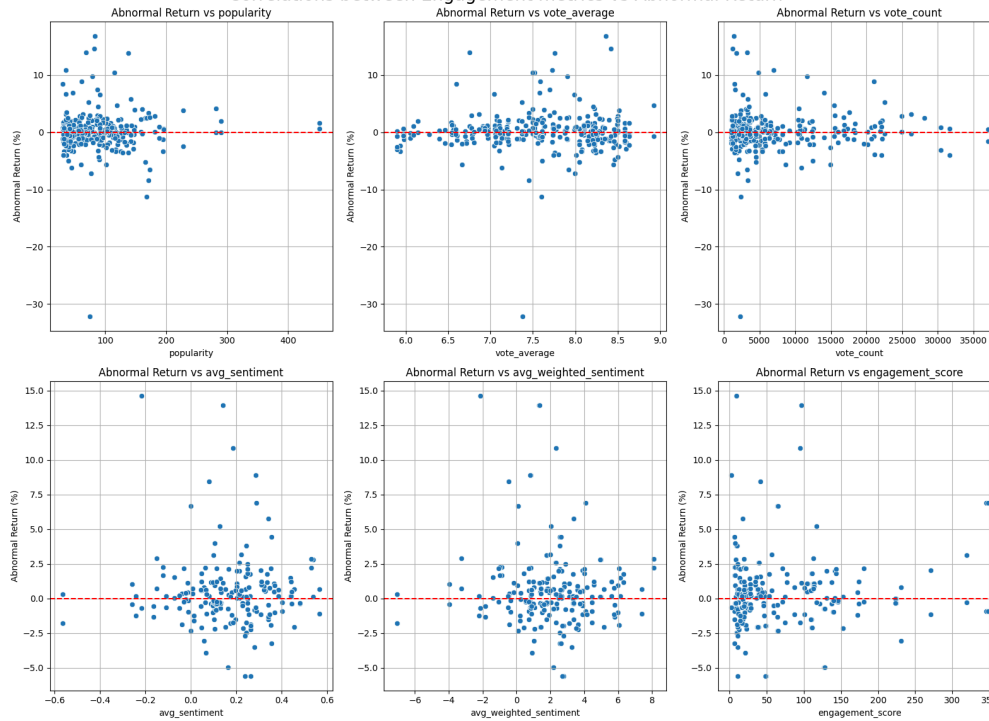
Netflix vs. Disney+ statistics:

t-statistic: 1.9156

p-value: 0.0558

A two-sample t-test comparing abnormal returns between Netflix and Disney+ yields a t-statistic of 1.92 and a p-value of 0.056. This result suggests a marginal difference in returns between platforms, approaching but not reaching conventional significance at the 5% level.

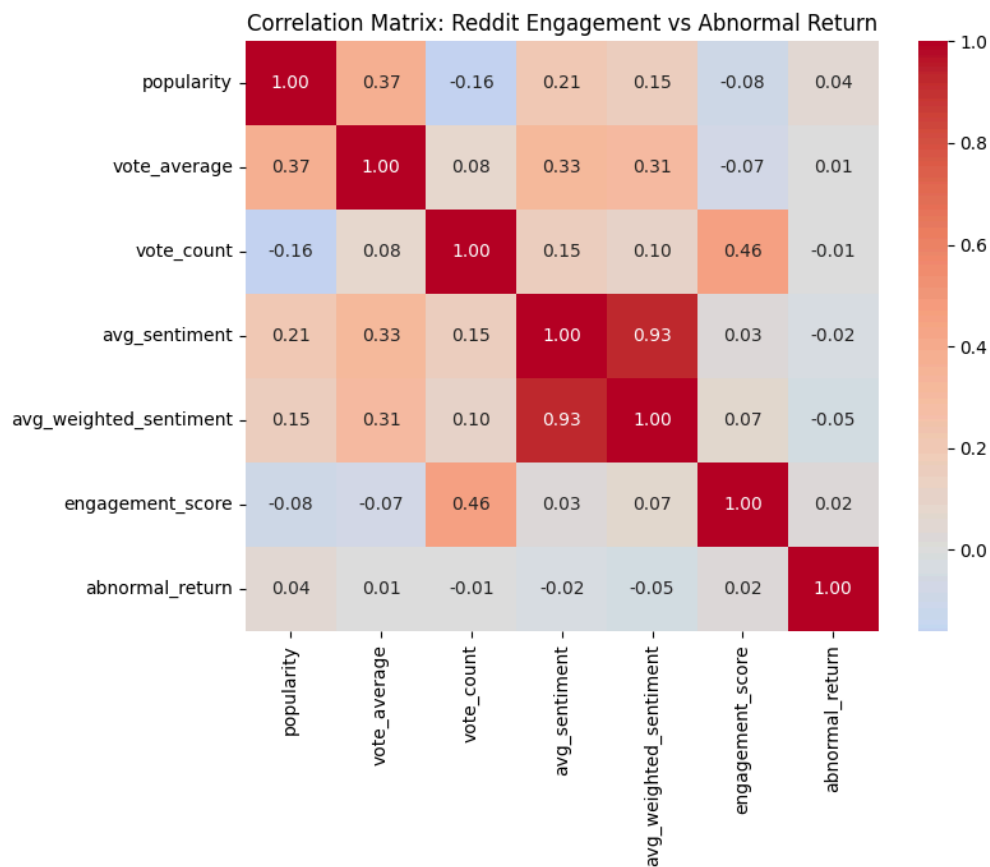
Correlations between Engagement metrics vs Abnormal Return



The scatterplots display the relationships between abnormal returns and various engagement-related metrics. Overall, the correlations appear weak, with most variables showing no clear linear pattern with abnormal return.

Engagement indicators such as popularity, vote_average, vote_count, and engagement_score show dispersed patterns with no discernible trend, suggesting limited explanatory power for short-term stock movement.

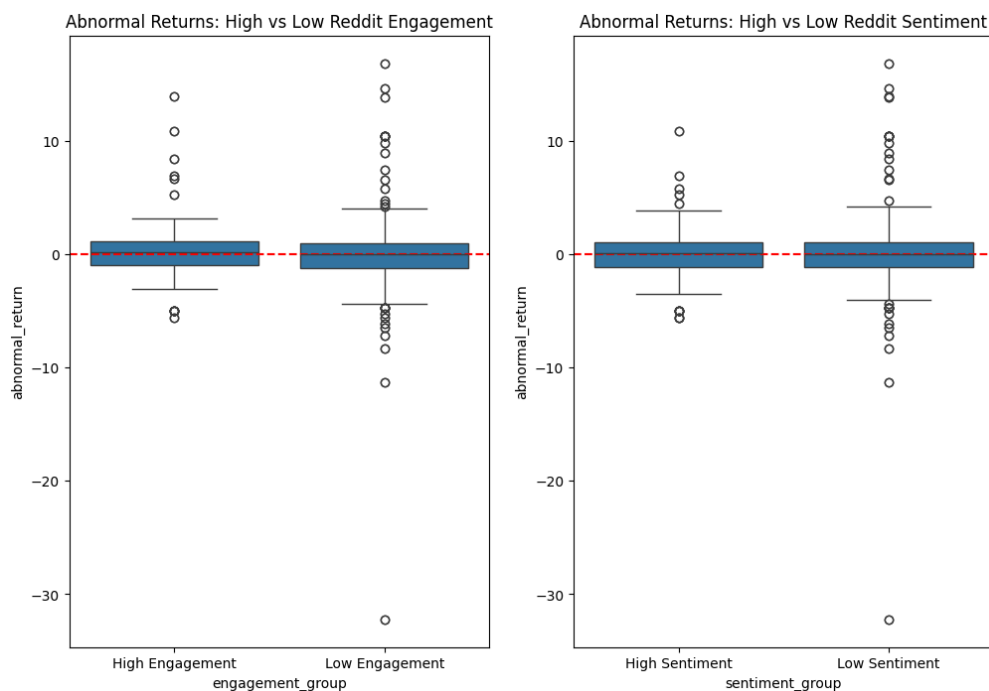
Sentiment variables (avg_sentiment and avg_weighted_sentiment) also display low association with abnormal returns, indicating that neither raw sentiment nor weighted sentiment is strongly linked to market reactions in the 3-day window following a release.



The correlation matrix further confirms the weak relationship between metrics of interest and the response variable (abnormal return). Most engagement and sentiment variables show near-zero correlation with abnormal returns, reinforcing the idea that these features have limited predictive power for short-term stock reactions.

These findings address our second subquestion, indicating that Reddit engagement metrics do not meaningfully correlate with stock price movements following major content releases.

Subquestion 3



Boxplots comparing abnormal returns between high and low Reddit engagement (left) and sentiment (right) groups reveal minimal differences in distribution. In both cases, the median abnormal return remains near zero, and the spread of values, including the presence of outliers, appears similar across groups—suggesting that neither high Reddit engagement nor sentiment is strongly associated with short-term stock price movement.

=== Engagement Groups ===

T-statistic: 0.9006

P-value: 0.3681

=== Sentiment Groups ===

T-statistic: -0.3355

P-value: 0.7373

The two-sample t-tests confirm that there is no statistically significant difference in abnormal returns between high and low Reddit engagement groups ($t = 0.900$, $p = 0.368$) or between high and low sentiment groups ($t = -0.335$, $p = 0.737$). These results reinforce the conclusion that neither engagement level nor sentiment strength meaningfully influences short-term stock performance following streaming releases.

OLS Regression Results

Dep. Variable:	abnormal_return	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.002
Method:	Least Squares	F-statistic:	0.1634
Date:	Tue, 29 Apr 2025	Prob (F-statistic):	0.686
Time:	22:33:25	Log-Likelihood:	-955.22
No. Observations:	402	AIC:	1914.
Df Residuals:	400	BIC:	1922.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.3633	0.173	2.098	0.036	0.023	0.704
avg_sentiment	-0.2865	0.709	-0.404	0.686	-1.680	1.107

Omnibus:	209.164	Durbin-Watson:	0.923
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1649.800
Skew:	2.082	Prob(JB):	0.00
Kurtosis:	12.009	Cond. No.	5.59

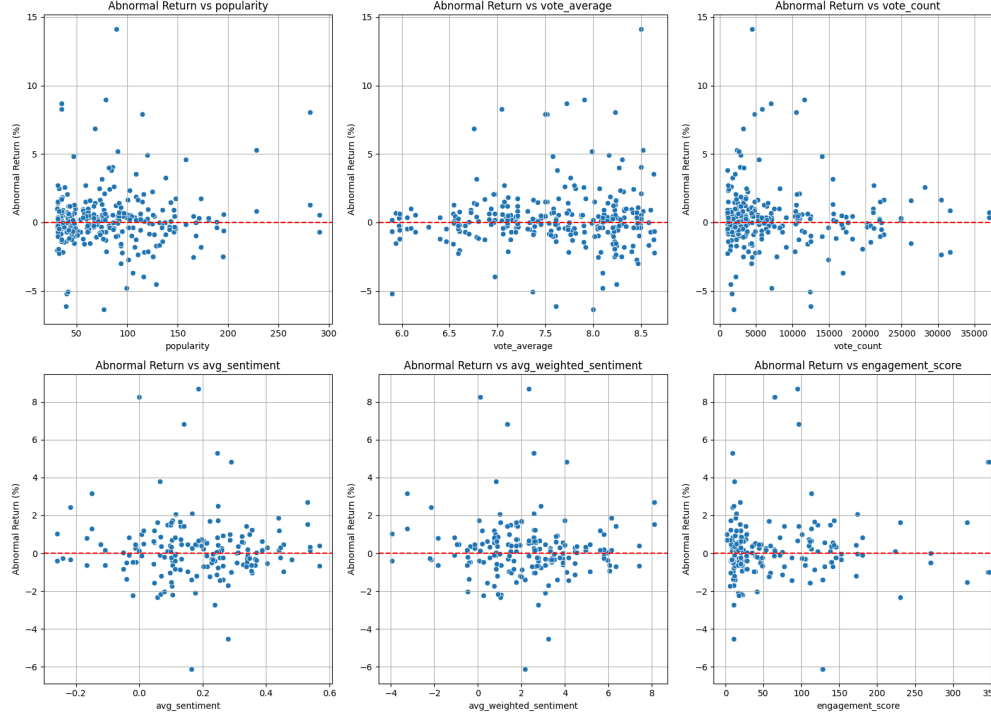
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

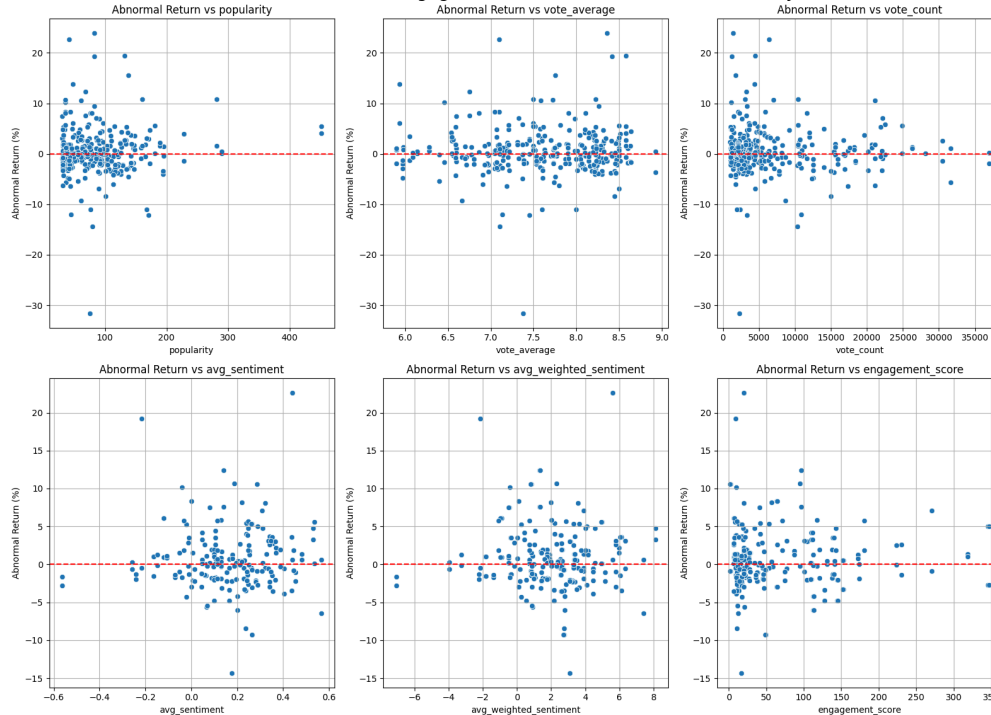
The linear regression analysis shows no statistically significant relationship between Reddit average sentiment and abnormal stock returns following Netflix or Disney+ content releases. The direction of the estimated relationship is negative, but the effect is tiny and not significant. In other words, based on regression analysis, we find no evidence that the sentiment of Reddit comments about a movie or show is associated with upward or downward movements in the stock price of Netflix or Disney+ in the short term. This finding directly addresses our third subquestion, providing no evidence of a meaningful association between online sentiment and stock performance in the short term.

Subquestion 4

Correlations between Engagement metrics vs Abnormal Return Day 1

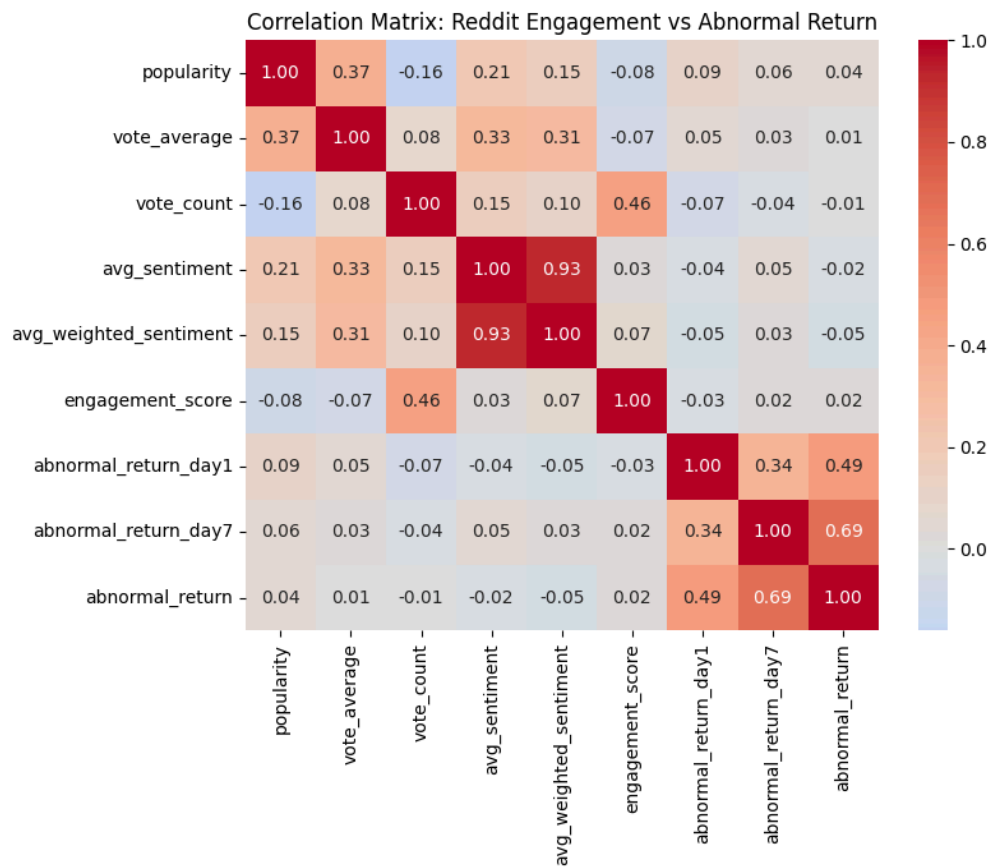


Correlations between Engagement metrics vs Abnormal Return Day 7



It is important to note that the abnormal returns for lag day 3 correspond exactly to the abnormal returns we have analyzed so far.

For other lag periods (1-day and 7-day), the correlation analysis reveals no visible relationships between Reddit engagement or sentiment metrics and the different types of abnormal returns.



The correlation matrix supports the conclusion that there is no meaningful lagged relationship between Reddit engagement or sentiment metrics and streaming providers' abnormal stock returns. Across all lag periods, correlations between engagement variables (e.g., engagement_score, avg_sentiment, avg_weighted_sentiment) and abnormal returns remain close to zero.

While there is a visible correlation between the movie's popularity with its sentiment, it is not useful to our main research question purpose.

Regression Model for Abnormal Return Day 1:

OLS Regression Results

Dep. Variable:	abnormal_return	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.002
Method:	Least Squares	F-statistic:	0.1634
Date:	Tue, 29 Apr 2025	Prob (F-statistic):	0.686
Time:	22:33:36	Log-Likelihood:	-955.22
No. Observations:	402	AIC:	1914.
Df Residuals:	400	BIC:	1922.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.3633	0.173	2.098	0.036	0.023	0.704
avg_sentiment	-0.2865	0.709	-0.404	0.686	-1.680	1.107

Omnibus:	209.164	Durbin-Watson:	0.923
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1649.800
Skew:	2.082	Prob(JB):	0.00
Kurtosis:	12.009	Cond. No.	5.59

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression Model for Abnormal Return Day 7:

OLS Regression Results

Dep. Variable:	abnormal_return_day7	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.002
Method:	Least Squares	F-statistic:	0.1590
Date:	Tue, 29 Apr 2025	Prob (F-statistic):	0.690
Time:	22:33:36	Log-Likelihood:	-1094.6
No. Observations:	386	AIC:	2193.
Df Residuals:	384	BIC:	2201.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.2875	0.298	0.966	0.335	-0.298	0.873
avg_sentiment	0.4929	1.236	0.399	0.690	-1.938	2.923

Omnibus:	139.727	Durbin-Watson:	0.913
Prob(Omnibus):	0.000	Jarque-Bera (JB):	805.029
Skew:	1.419	Prob(JB):	1.55e-175
Kurtosis:	9.481	Cond. No.	6.05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The linear regression analyses for both 1-day and 7-day abnormal returns show no statistically significant relationship with Reddit average sentiment. In both models, the R-squared value is effectively zero, indicating that average sentiment explains none of the variation in abnormal returns. The sentiment coefficient is small and statistically insignificant in each case (e.g., $p = 0.690$ for the 7-day model), suggesting no meaningful effect. Overall, these results reinforce that Reddit sentiment does not predict short-term stock movements, even when accounting for potential lagged effects.

Based on these results, we find no evidence of a lagged relationship between Reddit activity and the short-term stock performance of streaming providers, addressing our fourth subquestion.

Logistic regression model analysis

Looking at our previous results, it seems like linear model does not work well with the data. We may want to choose logistic regression as our predictive model instead because it's a natural fit for turning the event-study question "did the 3-day abnormal return go up or not?" into a binary classification problem. By modeling the log-odds of a positive return, logistic regression gives us easily interpretable coefficients that tell us how each feature shifts the probability of an uptick.

AUC: 0.4847222222222223

	precision	recall	f1-score	support
0	0.38	0.73	0.50	30
1	0.60	0.25	0.35	48
accuracy			0.44	78
macro avg	0.49	0.49	0.43	78
weighted avg	0.52	0.44	0.41	78

Confusion matrix:

```
[[22  8]
 [36 12]]
```

The logistic regression model essentially failed to distinguish days with positive versus non-positive 3-day abnormal returns: an AUC of 0.485 means it performs slightly worse than random chance, and the confusion matrix (22 true negatives vs. 8 false positives; 12 true positives vs. 36 false negatives) shows a strong bias toward predicting “down” days. Although it catches 73% of the actual down-moves, over two-thirds of up-moves are missed (25% recall on positives), and its low precision on negatives (38%) underscores that even its “down” predictions aren’t reliably correct.

Results and Discussion

1. Abnormal Returns Following Content Releases

- **Key Finding:**

No statistically significant abnormal returns were observed in the 3-day post-release window (mean = 0.36%, $t = 0.855$, $p = 0.393$).

- **Interpretation:**

- Supports the **efficient market hypothesis**—stock prices may already reflect content expectations pre-release.
- **Netflix** showed higher volatility than **Disney+**.

- **Contrast with Prior Work:**

Contradicts anecdotal reports of “hit-driven” stock bumps (e.g., *Stranger Things*), suggesting these are outliers.

2. Reddit Engagement and Stock Price Correlation

- **Key Metrics:**
 - Engagement score: $r = 0.02$ ($p = 0.68$).
 - Avg. sentiment: $r = -0.02$ ($p = 0.69$).
- **Visual Evidence:**

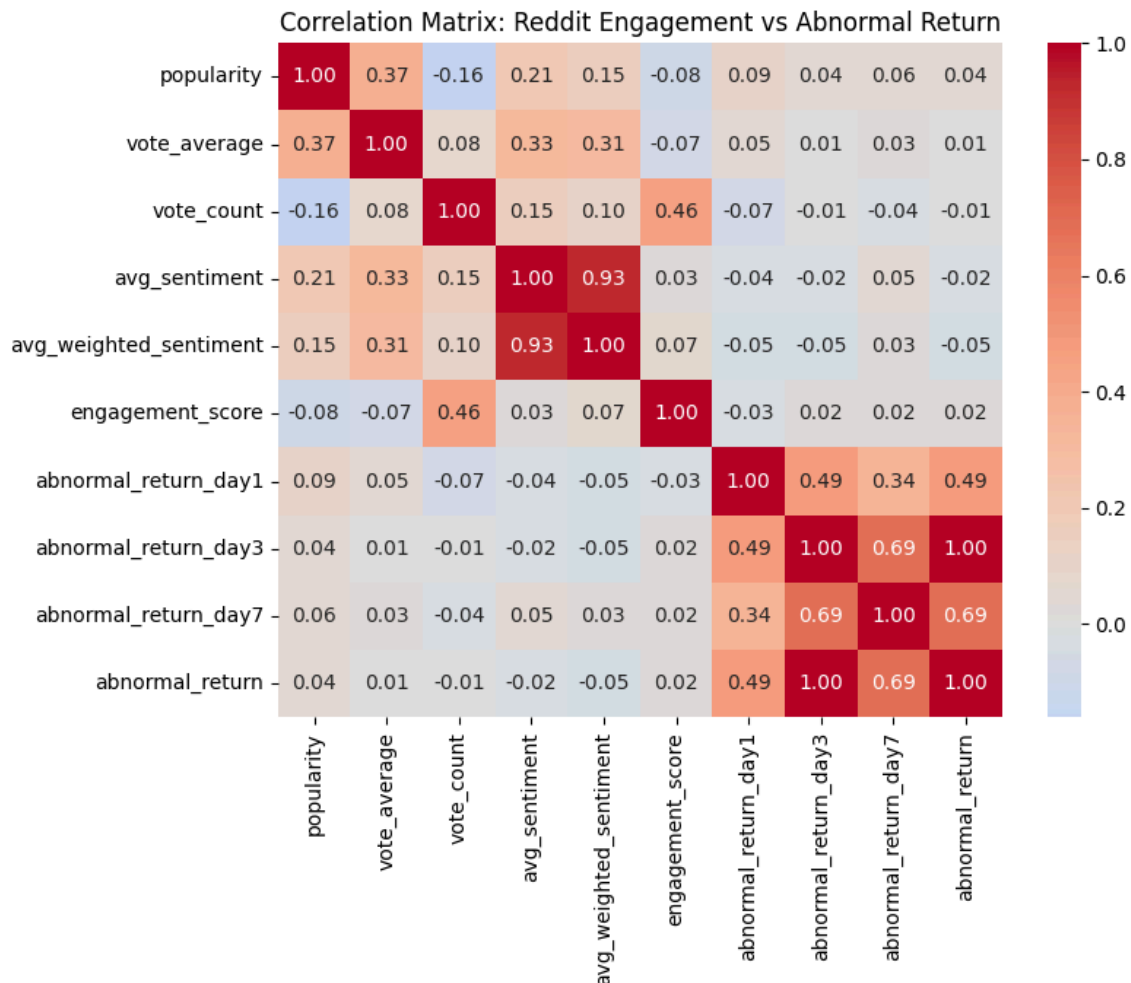


Figure 1: Heatmap showing near-zero correlations between engagement metrics and abnormal returns.

- **Why No Correlation?**
 - Reddit's niche audience $\neq$ investor demographics.
 - Engagement may affect **long-term subscriber growth** rather than short-term stocks.

3. Sentiment Analysis and Market Reactions

- **Regression Results:**
 - Weighted sentiment coefficient: -0.29% ($p = 0.686$, $R^2 \approx 0$).
 - High vs. low sentiment groups: $t = -0.336$ ($p = 0.737$).
- **Limitations:**
 - **VADER** may misclassify nuanced language (e.g., sarcasm).
 - Negative sentiment could reflect fan debates (e.g., *The Witcher*) rather than true disapproval.

4. Lagged Effects (1-Day, 3-Day, 7-Day Windows)

- **Results:**
 - All lagged correlations: $|r| < 0.05$.
 - No significant coefficients in lagged regressions ($p > 0.05$).
- **Takeaway:**

Reddit activity is **not a leading indicator**, even with delayed market windows.

5. Logistic Regression Analysis

- **Logistic Regression Results**

On the held-out 20 % test set ($n = 78$), our regularized logistic regression yielded the following performance:

AUC: 0.485

Accuracy: 0.44

Confusion matrix:

- True negatives (correct “down”): 22
- False positives (predicted “up” but actual “down”): 8
- False negatives (predicted “down” but actual “up”): 36
- True positives (correct “up”): 12

Class-wise precision / recall / f1-score:

- Down (0): precision = 0.38, recall = 0.73, f1 = 0.50
- Up (1): precision = 0.60, recall = 0.25, f1 = 0.35

These metrics indicate that the model is slightly worse than random at ranking events (AUC < 0.50) and biased toward predicting “down”, capturing most down-moves but missing 75% of up-moves. The low f1-scores and overall accuracy confirm that this baseline logistic regression provides no reliable signal for short-term abnormal return direction in its current form.

Theoretical and Practical Implications

For Investors:

- **Avoid** using Reddit buzz as a stock signal.
- Focus on **fundamentals** (subscriber growth, revenue).

For Platforms:

- Negative sentiment may signal creative risks (e.g., franchise fatigue).

For Researchers:

- Future work: Test **Twitter/X data** or **nonlinear models**.

Limitations

- Short-term window ignores long-term cultural impact.
- Selection bias (only high-popularity releases analyzed).
- External confounders (e.g., macroeconomic events) uncontrolled.

Conclusion

This study systematically examined the relationship between streaming content releases, Reddit engagement metrics, and short-term stock performance for Netflix and Disney+. The key takeaway is clear: **neither content releases nor social media buzz on Reddit significantly predict abnormal stock returns** in the 1-day, 3-day, or 7-day windows following a release. While this null result may seem counterintuitive given popular narratives about "hit-driven" stock surges, it aligns with broader financial theories and suggests nuanced realities about market behavior.

Future Research Directions

To build on these findings, future work could:

1. Extend the Time Horizon

- Test 30-/90-day windows to capture delayed effects (e.g., awards, word-of-mouth growth).

2. Incorporate Alternative Data

- **Twitter/X**: Broader demographic representation.
- **YouTube/Google Trends**: Measure viewership demand directly.

3. Control for Confounders

- Use multivariate models to account for earnings announcements, competitor releases, or macroeconomic trends.

Final Thoughts

While this study challenges the simplistic "good content → stock surge" narrative, it underscores the complexity of linking cultural phenomena to financial markets. For investors, the lesson is to **look beyond social media hype** and focus on fundamentals like subscriber economics and content ROI. For researchers, this highlights the need to improve methods by combining both numbers and real-world insights.